

Project Summary

This project focuses on developing a complete end-to-end data preprocessing and feature engineering pipeline for a Customer Credit Risk Analysis system. The primary objective is to prepare a high-quality, structured, and machine-learning-ready dataset that can be used to predict whether a customer is likely to default on a loan (`default_flag`).

The dataset used in this project contains detailed customer-level financial, transactional, and demographic information such as income, loan amount, credit score, transaction behavior, loan purpose, and personal attributes. To simulate real-world data engineering scenarios, the data is collected and integrated from multiple heterogeneous sources including CSV files, SQL databases, JSON files, and external APIs. This ensures that the pipeline is robust and capable of handling multi-source data integration challenges commonly seen in production systems.

Once the data is collected, a series of preprocessing steps are applied to ensure data quality and consistency. Missing values are handled using appropriate imputation strategies: numerical features are filled using mean or median values depending on distribution, while categorical features are handled using mode imputation. Outlier detection is performed using the Interquartile Range (IQR) method, and robust scaling techniques are applied to reduce the impact of extreme values.

Categorical variables such as gender, region, and loan purpose are transformed into numerical format using encoding techniques. One-hot encoding is applied for nominal variables, while label encoding is used where ordinal relationships exist. This ensures that machine learning models can effectively interpret categorical information.

Feature scaling is performed using multiple techniques including standardization (Z-score scaling), Min-Max normalization, Max-Abs scaling, and Robust scaling. These transformations ensure that all numerical features are brought to a comparable scale, improving model performance and convergence.

To address skewness and improve feature distribution, several mathematical transformations are applied, including logarithmic transformation, square root transformation, reciprocal transformation, and power transformations such as Box-Cox and Yeo-Johnson methods. These transformations help stabilize variance and make the data more suitable for modeling.

In addition to preprocessing, significant feature engineering is performed to enhance the predictive power of the dataset. New features are created such as Debt-to-Income Ratio, Spending-to-Income Ratio, and Average Monthly Transactions. Furthermore, unsupervised learning techniques like KMeans clustering are used to segment customers into income-based groups, adding additional behavioral insights to the dataset.

The final output of this project is a clean, fully transformed, and feature-engineered dataset that is free from missing values, inconsistencies, and extreme outliers. The dataset is structured and optimized for machine learning tasks, particularly for classification models aimed at predicting credit risk.

Overall, this project demonstrates a complete real-world data science workflow, covering data integration, preprocessing, transformation, and feature engineering, making it suitable for building reliable and production-ready credit risk prediction models.